Harshavardhini Jonnalagadda

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SUMMARY

Design Verification Engineer with hands-on experience in SystemVerilog, UVM, and RTL verification, building testbenches, debugging designs, and ensuring functional coverage.
Comfortable using Python and TCL for automation, with a strong foundation in computer architecture and verification concepts.

TECHNICAL SKILLS

Languages & HDLs: Verilog, SystemVerilog, VHDL, C, C++, Python, TCL ,perl..
Verification: UVM, Assertion-Based Verification, Functional Coverage, Constrained-Random Verification.
Formal Verification & Concepts: Basic Formal Logic, Intro to Model Checking, Intro to Static Analysis.
Protocols: AXI, AHB, APB, UART, SPI, I2C.
Tools: Synopsys VCS, Verdi, ModelSim, Vivado.
Concepts: RTL Design, FSM, Pipelining, Debugging, SRAM, DRAM,OOPS,SVA.
Environment: Linux.

EDUCATION

M.S. in Computer Engineering, California State University, Northridge, **Aug 2026**
B.Tech in Computer Engineering, Mahatma Gandhi University, **2023**

EXPERIENCE

Design Verification Trainee (Hands-on Industry Training), Sumedha IT Solutions (Dec 2022 - Dec 2023)

1. Developed **SystemVerilog testbenches** for RTL verification with focus on functional correctness and edge-case validation.
2. Applied **UVM methodology** using driver, monitor, and scoreboard for structured verification.
3. Performed **RTL debugging using waveform analysis** to identify and resolve functional mismatches.
4. Validated designs using **AXI/APB protocol concepts and assertion-based checks**
5. Used **Synopsys VCS and Verdi** for simulation, debugging, and waveform analysis
6. Automated workflows using **Python and TCL scripting** in a Linux environment.
7. Explored **basic formal verification concepts** and collaborated with team members to improve verification quality.
8.Applied knowledge of **CPU architecture concepts** including pipelining, hazard detection, and control flow during design verification tasks

PROJECTS

Parameterized ALU + Register File Datapath

1.Built this project to understand how **RTL datapaths and processor-style computation** work internally.
2.Designed a **parameterized 32-bit ALU, register file, datapath integration**, and verified functionality through simulations and waveform debugging.
3.Faced challenges while **connecting datapath modules and debugging incorrect ALU outputs**, which were resolved through waveform analysis.
4.Successfully achieved **stable arithmetic and logical operations with correct datapath** execution.
5.Tools used: Verilog, SystemVerilog, Icarus Verilog, GTKWave, Git/GitHub.

Synchronous FIFO Design & Verification

1.Built this project to understand **FIFO memory architecture and verification methodologies** used in RTL design.
2.Designed a parameterized synchronous FIFO with **overflow/underflow protection, SVA assertions,** and interface-based SystemVerilog verification.
3.Faced issues **with pointer handling and FIFO boundary conditions**, which were resolved through assertions and waveform debugging.
4.Successfully verified stable **FIFO read/write behavior with correct status-flag operation**.
5.Tools used: Verilog, SystemVerilog, SVA, GTKWave, Icarus Verilog, Git/GitHub.

COURSEWORK: Digital Design, Computer Architecture, VLSI Design, ASIC Development, System on Chip (SoC), Advanced Switching Theory, System Verilog, Digital System Structure, Machine Learning, Database Design.